

Information Field Theory

And the Operational Dissolution of the Hard Problem of Consciousness

IFT-Consciousness v2.0

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An operational formulation of consciousness is presented within Information Field Theory (IFT). Consciousness is defined as a regime of informational self-presence that emerges when a physical system simultaneously satisfies conditions of causal informational integration, temporal coherence, and embodied self-regulation. The functional Φ_{IFT} is introduced with three terms (Kullback–Leibler divergence with respect to the homeostatic basal state, Fisher curvature of the informational geometry, and irreducible causal information), together with a dual criterion of conscious presence ($\Phi_{\text{IFT}} > \Phi_c \wedge \tau \Xi > K$). A complete experimental program is developed with protocols for estimating $\rho(x, t)$ from EEG/fMRI data. The document incorporates 2024–2026 experimental data (Cogitate Consortium, empirical tests of IIT and GNWT, cerebellar studies) as partial validation of the predictions. The hard problem is operationally dissolved by transforming the metaphysical question into a measurable, falsifiable, and calibratable scientific program.

Contents

1	The Hard Problem and Its Dissolution	5
1.1	Classical Formulation	5
1.2	The Hidden Presuppositions	5
1.3	The Strategy of Operational Dissolution	5
1.4	Response to the Classical Arguments	6
2	IFT Foundations for Consciousness	7
2.1	Informational Core	7
2.2	Central Thesis	7
2.3	Seven Constitutive Conditions	7
3	Operational Definition of Consciousness	8
3.1	Dual Criterion	8
3.2	Formal Definition	8
4	The Consciousness Functional Φ_{IFT}	9
4.1	General Definition	9
4.2	Informational Term: KL Divergence	9
4.3	Geometric Term: Fisher Curvature	9
4.4	Causal Term: Irreducible Causal Information	9
5	Operationalization of $\rho(x, t)$ in Neural Systems	10
5.1	Estimation from High-Density EEG	10
5.2	Calculation of the Informational Gradient	10
5.3	Calculation of τ	10
6	IFT Neurobiology of Consciousness	11
6.1	Neurobiological Principle	11
6.2	Posterior Cortical Level	11
6.3	Cerebellum	11
7	States of Consciousness According to IFT	12
8	Qualia and Topology	13
8.1	Restricted Topological Hypothesis	13
8.2	Phenomenological Equivalence Principle	13
9	Artificial Intelligence and Consciousness	14
10	Experimental Program	15
10.1	Main Study Protocol	15
10.2	Preregistered Hypothesis	15
10.3	Anesthesia Protocol	15

11 Refutation Criteria	16
12 Comparison with Existing Theories	17
13 Master Formula of IFT-Consciousness	18
14 Final Declaration	19

Chapter 1

The Hard Problem and Its Dissolution

1.1 Classical Formulation

David Chalmers (1995) formulated the hard problem of consciousness, distinguishing it from the easy problems (cognitive functions explainable by physical mechanisms: perceptual discrimination, information integration, introspective access, verbal report, attentional control). The hard problem asks: why are certain physical processes accompanied by qualitative subjective experience? Why is there “something it is like” from the inside?

The classical arguments supporting the hard problem are:

1. **Philosophical zombie:** A being functionally identical to a human but without subjective experience is conceivable. If this is conceivable, experience is not reducible to functional properties.
2. **Mary’s argument (Jackson, 1982):** A scientist who knows all the physics of color but has never seen colors learns something new upon seeing red for the first time. There would be non-physical facts about experience.
3. **Explanatory gap (Levine, 1983):** Even if consciousness were identical to neural states, a gap would persist between functional description and phenomenal quality.

1.2 The Hidden Presuppositions

IFT identifies three questionable presuppositions in the formulation of the hard problem:

1. **Separability:** Functional structure and phenomenology are conceptually separable.
2. **Additivity:** Phenomenology is something added to functional processes, not constitutive of them.
3. **Complete propositional accessibility:** All genuine knowledge is expressible as propositions.

1.3 The Strategy of Operational Dissolution

IFT does not attempt to solve the hard problem by providing a mechanism that produces experience as output. It proposes a **dissolution**: to show that the question presupposes a separation between physical structure and phenomenology that does not exist in systems with sufficient embodied informational organization.

Definition 1.1 (Operational Dissolution). *To transform the question “why is there experience in addition to physics?” into a measurable problem about physical organization, causal integration, informational geometry, temporal coherence, and embodied dynamics.*

The distinction between functional structure and phenomenology reflects two modes of access to the same phenomenon:

Third person = physical-informational dynamics

First person = phenomenological self-presence

1.4 Response to the Classical Arguments

Response to the zombie: In systems with $\Phi_{\text{IFT}} > \Phi_c \wedge \tau\Xi > K$, causal informational integration and subjective self-presence are the same phenomenon. An IFT zombie would be a system that satisfies the conditions but lacks experience—an internal contradiction.

Response to Mary: What Mary learns is not a new propositional fact, but a new representational capacity: to instantiate the informational state corresponding to red in her own biological system.

Response to the explanatory gap: The gap reflects the absence of the correct description, not an ontological impossibility. If IFT succeeds in quantitatively predicting conscious contents from informational structure, the gap closes.

Chapter 2

IFT Foundations for Consciousness

2.1 Informational Core

IFT starts from a positive scalar field $\rho(x, t) > 0$ representing relational information density. The derived variables are:

$$u = \sqrt{\rho}, \quad \varphi = \ln \rho, \quad \Xi_\mu = \nabla_\mu \ln \rho, \quad \Xi = |\nabla \ln \rho|. \quad (2.1)$$

In the neural context, ρ is estimated from neuroimaging data (EEG, MEG, fMRI, ECoG), representing activity density, metabolic energy, or dynamic connectivity.

2.2 Central Thesis

Theorem 2.1 (IFT-Consciousness Thesis). *Consciousness is the internal perspective of an embodied informational dynamics that is sufficiently integrated, causal, coherent, and self-regulated. There are not two facts (physical process + added experience), but a single process described from two complementary perspectives.*

2.3 Seven Constitutive Conditions

IFT proposes that consciousness emerges when a physical system simultaneously satisfies:

1. **High informational integration:** The system integrates information as a whole.
2. **Intrinsic causality:** Causal dynamics emerge from within.
3. **Sustained temporal coherence:** Organization persists over time.
4. **Self-regulation (homeostasis):** The system maintains a basal reference state.
5. **Sufficient internal differentiation:** The system is not homogeneous.
6. **Sensorimotor or bodily coupling:** The system is embodied.
7. **Dynamic self-reference:** The system can represent itself.

Chapter 3

Operational Definition of Consciousness

3.1 Dual Criterion

Definition 3.1 (IFT Consciousness Criterion). *A physical system S is a candidate for consciousness if there exists a region Ω such that:*

$$\boxed{\Phi_{\text{IFT}}[S, \Omega] > \Phi_c \quad \wedge \quad \tau[S, \Omega] \Xi[S, \Omega] > K}, \quad (3.1)$$

where Φ_{IFT} is the causal informational integration functional, Φ_c is a critical threshold (to be empirically calibrated), τ is the temporal coherence time, $\Xi = |\nabla \ln \rho|$ is the informational gradient, and K is an informational coherence threshold.

This condition is an **operational strong hypothesis**, not a closed theorem. Its scientific value lies in its falsifiability.

3.2 Formal Definition

Definition 3.2 (Consciousness). *A regime of informational self-presence in which a physical system integrates information in a causally irreducible, temporally coherent, and self-regulated manner, generating a unified internal perspective.*

Chapter 4

The Consciousness Functional Φ_{IFT}

4.1 General Definition

$$\Phi_{\text{IFT}} = \int_{\Omega} dV w(x) [\alpha_I T_I(x) + \alpha_F T_F(x) + \alpha_C T_C(x)], \quad (4.1)$$

with normalized weights $\alpha_I + \alpha_F + \alpha_C = 1$, $\alpha_i \geq 0$, and anatomical or functional window $w(x)$.

4.2 Informational Term: KL Divergence

$$T_I(x) = \frac{\rho(x) \ln(\rho(x)/\rho_{\text{base}}(x))}{\rho_*}. \quad (4.2)$$

Measures informational deviation from the homeostatic basal state ρ_{base} (estimable during rest, anesthesia, or deep sleep).

4.3 Geometric Term: Fisher Curvature

$$T_F(x) = \ell_F^2 \text{Tr}(\kappa_F(x)), \quad (4.3)$$

where κ_F is the Ricci curvature associated with the Fisher metric induced by ρ on the state space.

4.4 Causal Term: Irreducible Causal Information

$$T_C(x) = I_{\text{causal}}(x), \quad (4.4)$$

which measures how much information the system integrates as a whole beyond the maximum information of its parts:

$$I_{\text{causal}}(\Omega) = \text{MI}(\Omega^{\text{future}}; \Omega^{\text{past}}) - \max_{\text{partitions } P} \sum_k \text{MI}(P_k^{\text{future}}; P_k^{\text{past}}). \quad (4.5)$$

Chapter 5

Operationalization of $\rho(x, t)$ in Neural Systems

5.1 Estimation from High-Density EEG

$$\rho_{\text{EEG}}(x_i, t) = \frac{P(x_i, t) + \epsilon}{\sum_{j=1}^{N_e} (P(x_j, t) + \epsilon)}, \quad (5.1)$$

where $P(x_i, t) = \int_{f_{\min}}^{f_{\max}} |\text{FFT}(V(x_i, t'))|^2 df$ over a window Δt .

5.2 Calculation of the Informational Gradient

$$\Xi_i(t) = \left| \sum_j A_{ij} [\ln \rho_j(t) - \ln \rho_i(t)] \right|, \quad (5.2)$$

with A_{ij} the connectivity matrix.

5.3 Calculation of τ

$$C_i(\Delta t) = \langle \rho_i(t) \rho_i(t + \Delta t) \rangle \sim e^{-\Delta t / \tau_i}. \quad (5.3)$$

Chapter 6

IFT Neurobiology of Consciousness

6.1 Neurobiological Principle

Biological consciousness does not arise from a single region but from a dynamic organization among multiple levels: brainstem and arousal systems, thalamus (integration regulator), hypothalamus (homeostasis), insula (interoception, bodily self-presence), cingulate cortex (motivation, valence), posterior cortex (perceptual content), frontoparietal networks (access, control, report), hippocampus (episodic memory, narrative continuity), and autonomic systems and body.

6.2 Posterior Cortical Level

IFT predicts:

$$\Phi_{\text{IFT}}(\text{posterior}) > \Phi_{\text{IFT}}(\text{frontal}) \quad (6.1)$$

for perceptual conscious content, consistent with the Cogitate Consortium (2025).

6.3 Cerebellum

High processing capacity but low global phenomenological integration. IFT predicts:

$$\Phi_{\text{IFT}}^{\text{cerebellum}} < \Phi_{\text{IFT}}^{\text{posterior cortex}}, \quad (6.2)$$

validated by recent studies (2025).

Chapter 7

States of Consciousness According to IFT

State	Φ_{IFT}	$\tau\Xi$	Consciousness
Alert wakefulness	$\gg \Phi_c$	$\gg K$	Full
REM sleep	$\approx \Phi_c$ or $> \Phi_c$	$\approx K$	Oneiric
Deep NREM sleep	$< \Phi_c$	$< K$	Minimal or absent
Deep general anesthesia	$< \Phi_c$	$< K$	Absent
Vegetative state	Fragmented, low	Low	Typically absent
Psychedelics	Curvature $\uparrow$	Variable	Intense, reduced control
Deep meditation	Stable, $> \Phi_c$	$> K$	Expanded
Current digital AI	Possibly computationally high	Probably $< K$	Probably not

Chapter 8

Qualia and Topology

8.1 Restricted Topological Hypothesis

IFT introduces as a secondary hypothesis that certain discrete phenomenological contrasts may correspond to stable topological classes of the phase $\Phi(x)$:

$$n_\gamma = \frac{1}{2\pi} \oint_\gamma \nabla \Phi \cdot d\mathbf{l} \in \mathbb{Z}. \quad (8.1)$$

8.2 Phenomenological Equivalence Principle

$\Phi_{\text{IFT}}[S_1] = \Phi_{\text{IFT}}[S_2] \wedge \text{identical phase topology} \Rightarrow S_1 \text{ and } S_2 \text{ are phenomenologically indistinguishable.}$
(8.2)

Chapter 9

Artificial Intelligence and Consciousness

IFT does not assert “silicon $\Rightarrow$ no consciousness.” It asserts: “without real intrinsic causality, without real homeostasis, without real multiscale coherence $\Rightarrow$ probably no consciousness.” An artificial system would be a candidate if it implements: a proper body, energetic self-regulation, internal variables with valence, multiscale recurrent architecture, intrinsic causality, continuous sensorimotor integration, and Φ_{IFT} measurably above threshold.

Chapter 10

Experimental Program

10.1 Main Study Protocol

- **Participants:** $N \geq 256$.
- **Conditions:** Wakefulness, NREM/REM sleep, sedation, anesthesia, bistable perceptual tasks, trial-by-trial subjective report.
- **Measurements:** EEG, MEG, fMRI, ECoG (if available), autonomic physiology.

10.2 Preregistered Hypothesis

$$r(\Phi_{\text{IFT}}, \text{reported conscious intensity}) > 0.7. \quad (10.1)$$

Failure criterion: $r < 0.3$ in multisite studies.

10.3 Anesthesia Protocol

Measure Φ_{IFT} and $\tau\Xi$ before, during, and after loss of consciousness. Different anesthetics should affect different components.

Chapter 11

Refutation Criteria

1. **Strong neuroscientific refutation:** $r(\Phi_{\text{IFT}}, \text{report}) < 0.3$ in multisite studies.
2. **Anesthesia refutation:** $\Phi_{\text{IFT}} > \Phi_c$ and $\tau\Xi > K$ during deep anesthesia without reported experience.
3. **Low-integration refutation:** Rich experience with $\Phi_{\text{IFT}} \approx 0$.
4. **Topological qualia refutation:** No topological structure in perceptual transitions.

Chapter 12

Comparison with Existing Theories

IFT unifies aspects of IIT (integration) and GNWT (access/report) in a single framework, adding temporal coherence ($\tau\Xi$) and embodiment. Unlike IIT, IFT does not take Φ as an ontological primitive but derives it from the informational field ρ .

Chapter 13

Master Formula of IFT-Consciousness

$$\boxed{\Phi_{\text{IFT}} = \int_{\Omega} dV w(x) \left[\alpha_I \frac{\rho \ln(\rho/\rho_{\text{base}})}{\rho_*} + \alpha_F \ell_F^2 \text{Tr}(\kappa_F) + \alpha_C I_{\text{causal}} \right]}, \quad (13.1)$$

with criterion:

$$\boxed{\Phi_{\text{IFT}} > \Phi_c \quad \wedge \quad \tau \Xi > K}. \quad (13.2)$$

Chapter 14

Final Declaration

IFT-Consciousness proposes that consciousness is a physical-informational regime of embodied self-presence. It is not treated as an extra substance, epiphenomenon, or illusion, but as the internal perspective of a physical dynamics that achieves sufficient integration, intrinsic causality, temporal coherence, and self-regulation.

The hard problem is operationally dissolved: the question “why is there experience in addition to physics?” presupposes a separation that IFT denies in systems with sufficient embodied informational integration.

consciousness = embodied informational self-presence
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hard problem = ill-posed problem if phenomenology is separated from physical organization
